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Spectral Integrated Gradients for Coarse-to-Fine Feature Attribution

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TL;DR

Spectral IG (SIG) builds the IG integration path from the **SVD of the baseline→input difference**. Activating singular components from *largest to smallest* reveals **coarse before fine detail**, suppressing high-frequency gradient noise for **cleaner, more faithful** attributions at only ∼**16% overhead** over IG.

Motivation

- **Integrated Gradients (IG)** is axiomatic, but attribution quality hinges on the **integration path**.
- IG's **straight line** $\gamma(t)=x'+t(x-x')$ introduces *all* features uniformly, accumulating **high-frequency noise** ⇒ speckled maps.
- Prior fixes (GIG greedy, BIG blur, MIG/EIG VAE manifold) help but need per-step gradient search or a trained generative model.

Signal-to-Noise in Linear Paths

The linear path scales the whole spectrum by one scalar, injecting dominant structure $\sigma_1 u_1 v_1^\top$ and fine detail $\sigma_R u_R v_R^\top$ at the **same rate**. Early high-frequency noise triggers erratic gradients that accumulate into the map.

Why the SVD Path is Optimal

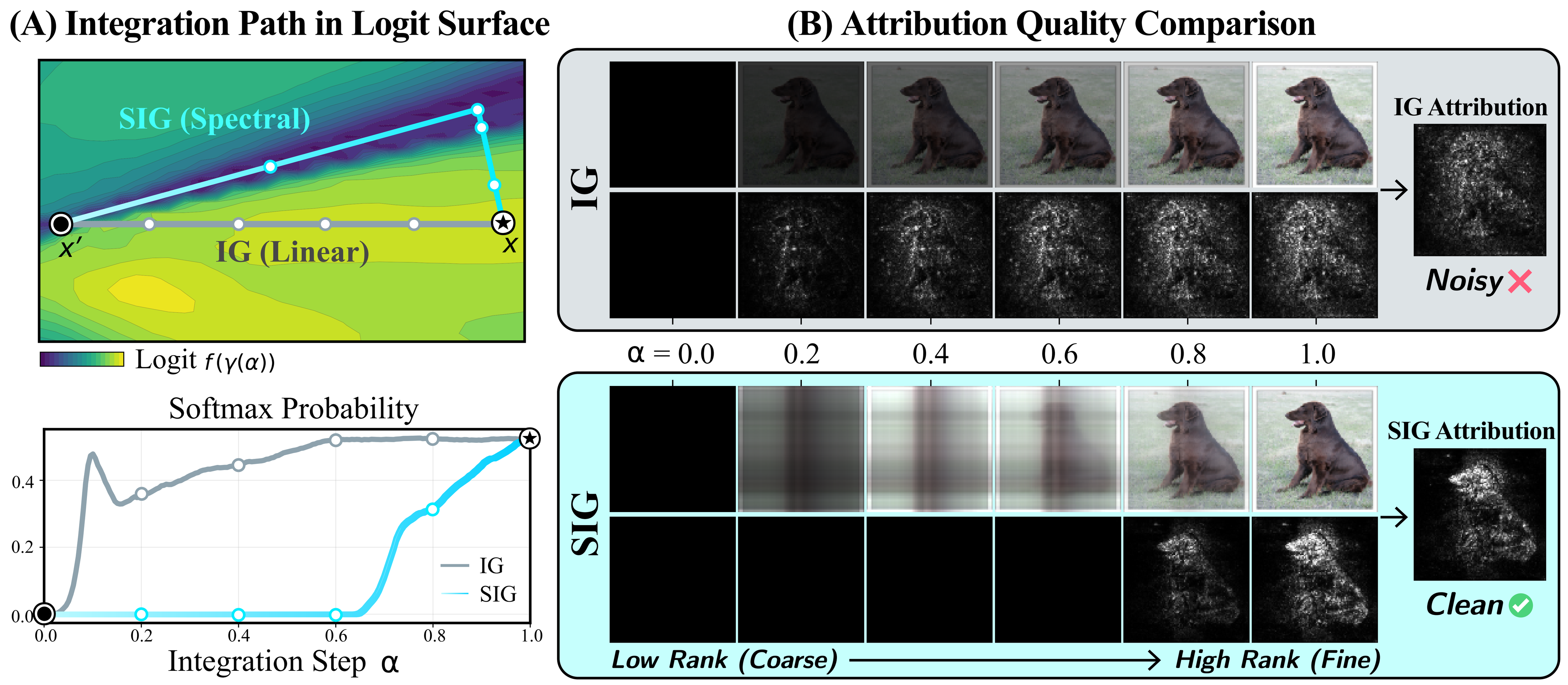
Coarse-to-fine is a **rank-constrained** problem with budget $k(\alpha)$.

Theorem (Eckart–Young–Mirsky). The best rank-k approximation of $\Delta=x-x'$ is its **truncated SVD**:

$$\sum_{i=1}^{k}\sigma_i u_i v_i^\top = \underset{\mathrm{rank}(\hat\Delta)\le k}{\operatorname{argmin}} \|\Delta-\hat\Delta\|_F .$$

So activating components *largest→smallest* is the **least-squares-optimal** coarse-to-fine path — **principled**, unlike heuristics (GIG, BIG, SAMP).

Idea: A Coarse-to-Fine Spectral Path



Overview of SIG. (A) On the logit landscape, **Linear IG** (gray) crosses unstable regions; **SIG** (cyan) prioritizes coarse (low-rank) before fine (high-rank) structure. **(B)** This suppresses high-frequency noise, giving cleaner attributions. The path runs from baseline ◉ to input ⊛.

Key idea. SVD ranks $\Delta=x-x'$: large singular values capture *dominant structure*, small ones *fine texture & noise*. Activating them large→small yields a **principled** coarse-to-fine path — provably optimal (see **Theorem**, left).

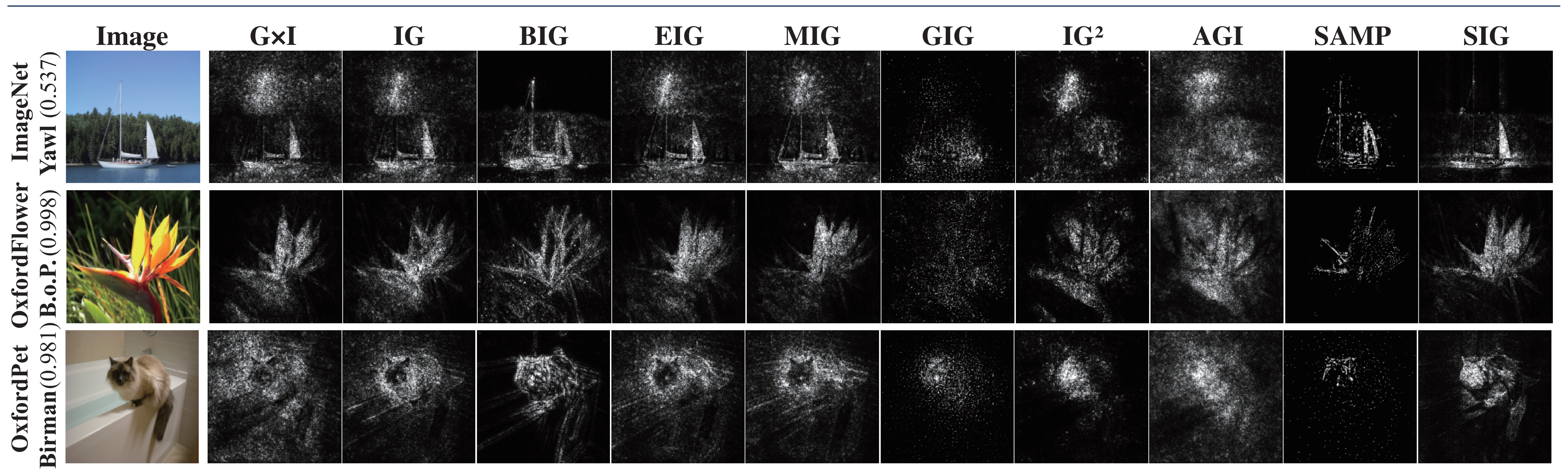
The SIG Path

A smooth gate $\phi_i(\alpha;\omega)\in[0,1]$ activates larger singular values *earlier*, making the truncated-SVD path continuous:

$$\gamma_{\mathrm{SIG}}(\alpha)=x'+\sum_{i=1}^{R}\phi_i(\alpha;\omega)\,\sigma_i u_i v_i^\top .$$

The overlap ω **interpolates**: $\omega\to 0$ = sequential SVD path; $\omega\to 1$ = **linear IG** ($\omega=0.4$), so SIG *generalizes* IG. **One SVD/image** ⇒ negligible overhead, while **keeping IG's axioms** + geometric symmetry.

Qualitative Comparison



ImageNet/InceptionV1, Oxford Pet/ResNet18, Flower/VGG16. SIG gives the **cleanest, sparsest** maps, suppressing background clutter and speckle (cf. SAMP).

Faithfulness on ImageNet

	ResNet18			VGG16			InceptionV1		
Method	DiffID ↑	Ins ↑	Del ↓	DiffID ↑	Ins ↑	Del ↓	DiffID ↑	Ins ↑	Del ↓
G×I	.104	.228	.124	.188	.275	.087	.138	.278	.140
IG	.159	.256	.097	.240	.309	.069	.192	.307	.116
BIG	<u>.312</u>	<u>.350</u>	<u>.038</u>	<u>.326</u>	.355	**.029**	<u>.347</u>	<u>.396</u>	<u>.048</u>
EIG	.150	.253	.103	.233	.305	.072	.191	.309	.118
MIG	.160	.256	.096	.235	.304	.069	.194	.310	.116
GIG	.251	.314	.064	.281	.330	.049	.303	.379	.076
AGI	.164	.255	.091	.226	.292	.066	.217	.310	.092
IG²	.182	.268	.085	.305	<u>.358</u>	.054	.257	.345	.088
SAMP	.288	.328	.040	.281	.312	<u>.031</u>	.309	.375	.066
SIG	**.342**	**.380**	**.038**	**.374**	**.411**	.037	**.378**	**.426**	**.048**

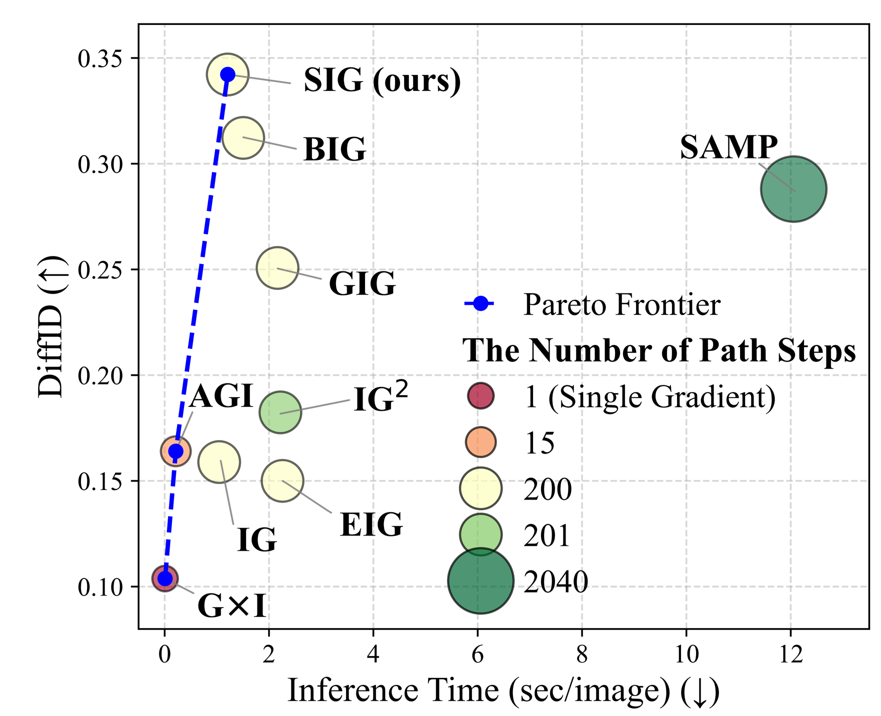
Best/<u>2nd</u>. Ins = Insertion Game, Del = Deletion Game. SIG best in nearly all settings.

Beyond Perturbation Metrics. Faithfulness holds across complementary protocols:

- **Leakage-aware ROAD:** best GAP on all 3 backbones (ResNet18 .267, VGG16 .301, IncV3 .310) — beating BIG (.252, .277, .303) everywhere.
- **Non-perturbation localization:** best mean BBox IoU, best Pointing Game (.669 vs. .648 for BIG), and best SAM-2 IoU (.232 vs. .217).

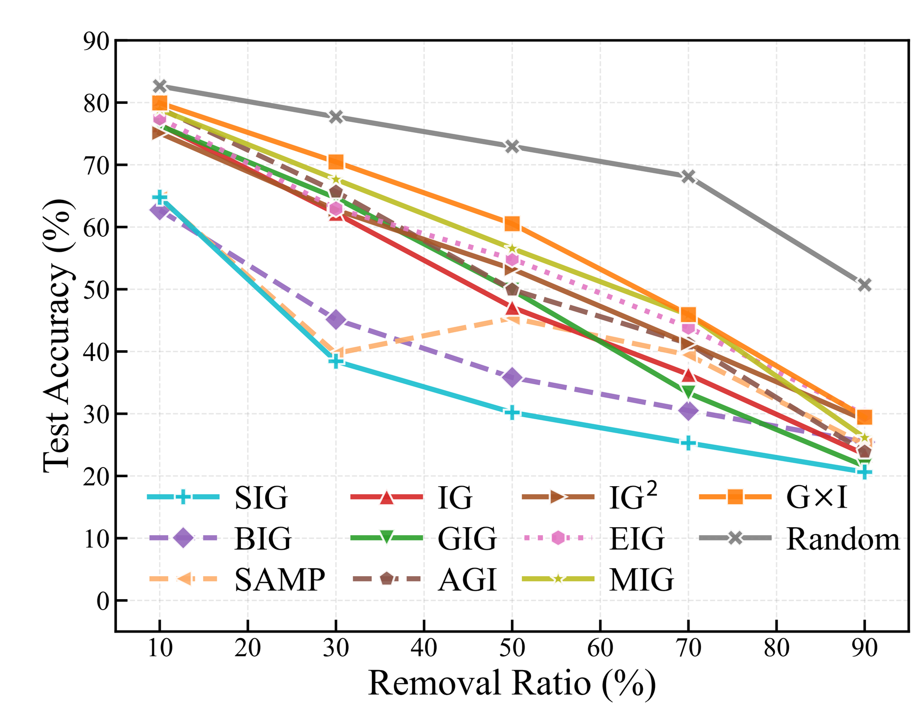
Efficiency & Robustness

Efficiency: Pareto-Optimal



DiffID vs. inference time (ImageNet/ResNet18) — SIG is **Pareto-optimal**: best DiffID at **1.21 s/img** (+16% vs. IG, vs. SAMP ∼12 s).

Robustness: ROAR on CIFAR-10



SIG (cyan) drops accuracy fastest under remove-and-retrain ⇒ **lowest AUC**, pinpointing the most discriminative pixels.

Take-aways

- A **single SVD** makes IG's path coarse-to-fine — **no generative model** (MIG, EIG, and MA-GIG), **no per-step search** (GIG, IG², AGI, and SAMP).
- **SOTA faithfulness** (also best leakage-aware ROAD & localization)
- while **preserving IG's axioms** at near-IG cost (1.21 s/img).